

A finite transfer cutoff for Collatz convergence

Collatz Lean research project

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Abstract

We quantify the existing affine-transfer criterion: instances with $36v + 27 \leq 3^b$ suffice for convergence of all positive seeds below 2^b . This supplies a justified convergence threshold from smaller transfer hypotheses and a conditional descent rule. The required transfers and global descent coverage remain unproved.

1 The cutoff theorem

Let $T(n) = n/2$ for even n and $(3n + 1)/2$ for odd n . Write $R(n)$ for eventual arrival at one and

$$A(v) : R(3v + 2) \implies R(27v + 20).$$

For every natural b , Lean proves

$$[\forall v, 36v + 27 \leq 3^b \implies A(v)] \implies [\forall n, 0 < n < 2^b \implies R(n)]. \quad (1)$$

More precisely, it suffices to assume only transfer instances arising from odd runs below the seed bound. The interval in (1) is a sufficient overestimate of those requests.

2 Proof mechanism

Use strong induction on $n < 2^b$. Even seeds and odd seeds with only one initial odd step descend directly. For the remaining odd seeds, write $n + 1 = 2^a m$ with m odd and $a \geq 2$. The classical double-even exit merges with $n - 1$, whose convergence is available by induction.

In the single-even exit, put $z = T^{a+2}(n - 1)$. The existing odd-run identities give $z \equiv 2 \pmod{3}$, so $z = 3v + 2$, and

$$12z + 3 = 3^a m, \quad 36v + 27 = 3^a m. \quad (2)$$

The target endpoint is $9z + 2 = 27v + 20$. Moreover, $2^a m \leq 2^b$ implies $a \leq b$ and $m \leq 2^{b-a} \leq 3^{b-a}$, hence $3^a m \leq 3^b$. Thus (2) places the required transfer inside the stated cutoff. Apply it to the convergent predecessor endpoint and transport back along the target orbit. No additional convergence premise enters this proof.

3 A sound descent rule

Suppose $A(v)$ is known for every $v < u$. If $3^b \leq 36u$, then $36v + 27 \leq 3^b$ implies $v < u$. Consequently (1) proves convergence of all positive seeds below 2^b . In particular,

$$\left. \begin{array}{l} \forall v < u, A(v), \quad 3^b \leq 36u, \\ T^t(27u + 20) < 2^b \end{array} \right\} \implies A(u). \quad (3)$$

The source convergence premise in $A(u)$ is not needed once these stronger descent hypotheses hold. A proof that (3) or another rule covers every parameter is still missing.

4 Validation and scope

The six declarations in `Collatz.BoundedTransfer` formalize the exact-request criterion, the size bounds, (1), its smaller-transfer corollary, and (3). The arguments use standard foundations; no conjectural transfer instance is imported as a fact.

For Mersenne parameters $u = 16 \cdot 64^h - 1$, the known reduced target is $729 \cdot 81^h - 10$. Choosing the largest integer b with $3^b \leq 36u$, exact Python searches find a descent below 2^b for every $h = 0, \dots, 512$. The maximum observed exit depth is 6268 at $h = 487$. These are finite computational witnesses, not Lean-certified family coverage. They support testing (3), not a global convergence claim. No literature-priority claim is made for this quantitative formulation.